

Symmetry and Groups — Lecture 1

IMAPP lecture notes

Spring 2026

1 Groups

A group G is a set equipped with an internal composition law, written multiplicatively, such that the following axioms hold:

- i. **Identity:** there exists $e \in G$ such that $eg = ge = g$ for every $g \in G$.
- ii. **Inverse:** for every $g \in G$ there is $g^{-1} \in G$ with $gg^{-1} = g^{-1}g = e$.
- iii. **Associativity:** $(g_1g_2)g_3 = g_1(g_2g_3)$ for all $g_1, g_2, g_3 \in G$.

2 Types of groups

- A **finite group** has finitely many elements. Its order is $|G|$; for example, the permutation group S_n has $|S_n| = n!$.
- A **discrete infinite group** has infinitely many but countably many elements, such as $(\mathbb{Z}, +)$.
- A **continuous group** has elements parametrized by one or more continuous parameters. The number of independent parameters is the dimension of the group; rotations provide a standard example.

3 Related definitions

A semigroup has a closed associative composition law but need not contain an identity or inverses. For instance, $(\mathbb{N}^*, +)$ is a semigroup.

The commutator of two elements is

$$[g_1, g_2] = g_1g_2g_1^{-1}g_2^{-1}.$$

The elements commute when $g_1g_2 = g_2g_1$. A group is **Abelian** precisely when all its elements commute; $(\mathbb{Z}, +)$ is Abelian.

For a finite group, the order of an element g is the smallest positive integer n satisfying

$$\text{ord}(g) = n, \quad g^n = e.$$

4 Cyclic groups

A cyclic group is generated by the powers of a single element:

$$C_n = \langle g \rangle = \{e, g, g^2, \dots, g^{n-1}\}, \quad g^n = e.$$

Every cyclic group is Abelian. Geometrically, C_n can be realized as the rotations about a fixed point through angles $2\pi k/n$.

5 Multiplication tables

For a finite group, the multiplication table records the complete group structure. The rearrangement property says that multiplying every element on the left or right by a fixed $g_0 \in G$ permutes the elements of G . Consequently, every element appears exactly once in every row and every column.

6 The symmetric group

S_n is the group of all permutations of an n -element set. A permutation is a bijection of the set to itself, and the group operation is composition. For example, in S_3 ,

$$(12)(23) = (123), \quad (23)(12) = (132),$$

so the order of composition matters: S_3 is non-Abelian.

Summary

Groups encode reversible composition laws. The first useful examples are cyclic groups and permutation groups; multiplication tables and element orders make their structure explicit.